COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
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ACTIVITY NO. **CPE 010-CPE21S4 - Data Structures and Algorithms**

1. PROCEDURE OUTPUT

Instructions:

- Please make sure to follow all code of conduct re: examinations and related policies.
- Students caught cheating **will not receive a grade for lab exam**.

Problem Title: The Customer Service Verification Problem

A customer service center processes customers in the order they arrive. Each customer is assigned a unique customer number.

At the end of the day, the supervisor wants to verify whether the order in which the customers were served is the **reverse of the order in which they originally arrived**.

Given a sequence of customer numbers, your task is to determine whether the sequence is the same when its order is reversed.

For example:

101 102 103 102 101

is a valid reverse-order sequence because the values read from the beginning and from the end are the same.

Your task is to use **two data structures** for your solution:

- Stack
- Queue

However, the two data structures **must have different internal representations**, either:

- Array-based
- Linked List

Example: If your stack is array-based, your queue must be implemented using a linked list, and vice versa.

Each customer number must be inserted into **both the Stack and Queue**.

You must then use the behavior of the two data structures to determine whether the sequence satisfies the required condition.

You may only use the **public operations provided by the existing Stack and Queue header files**. Do not modify the internal representation of either data structure.

The user is then prompted whether the given customer sequence satisfies the reverse-order condition.

Pseudocode

```
START

INPUT the customer numbers


        CREATE an empty STACK
        CREATE an empty QUEUE

FOR each customer number's individual numbers

        PUSH the customer number's individual value into the STACK
        ENQUEUE the customer number's individual value into the QUEUE

END FOR LOOP

WHILE the STACK is not empty

        GET one value by POPPING from the STACK
        GET one value by DEQUEUING from the QUEUE

BOOL  isEmpty

   RETURN (TOP == -1)

BOOL isFull

   RETURN (TOP == maxSize - 1)

FOR INT customer number

        IF Top isEmpty
                PUSH customer number

         ELSE

                  POP customer number
```

END WHILE **IF all values matched** **The sequence satisfies the reverse-order condition** **END IF** **STOP**

Screenshot of the code	

Array Implementation in stack

skills exam > C stackarr.h > pop()

```
1   #ifndef STACKARR_H
2   #define STACKARR_H
3
4   #include <iostream>
5
6   const int maxSize = 10;
7   int stackArr[maxSize];
8   int top = -1;
9
10  // observers
11  bool isFull();
12  bool isEmpty();
13  void Top();
14
15  // transformers
16  void pop();
17  void push(int newData); // to accept integer parameter
18
19
20  bool isEmpty()
21  {
22      return (top == -1);
23  }
24
25  bool isFull()
26  {
27      return (top == maxSize - 1);
28  }
29
30  void Top()
31  {
32      if (isEmpty())
33      {
34          std::cout << "Stack is Empty" << std::endl;
35          return;
36      }
37
38      std::cout << "The top element is: "
39                << stackArr[top]
40                << std::endl;
41  }
42
43  void push(int newData)
44  {
45      // error catching
46      if (isFull())
47      {
48          std::cout << "Stack Overflow!" << std::endl;
```

```
        return;
    }

    // storing to the stack
    stackArr[++top] = newData;
}

void pop()
{
    // error catching
    if (isEmpty())
    {
        std::cout << "Stack Underflow!" << std::endl;
        return;
    }

    // to remove the element from the stack
    top--;
}

#endif
```

Linked List Implementation in Queue

```
#ifndef QUEUELL_H
#define QUEUELL_H
#include <iostream>

// create a class node
template <typename T>
class qNode
{
public:
    T data;
    qNode *next;
};

// creating the new node
template <typename T>
qNode<T> *new_node(T newData)
{

    // allocate a space for the new node
    qNode<T> *newNode = new qNode<T>;
    // store the newData to the newNode
    newNode->data = newData;
    // point the newNode next to null
    newNode->next = nullptr;
    return newNode;
}

// enqueue
template <typename T>
void Enqueue(T newData, qNode<T> **frontPtr, qNode<T> **backPtr)
{

    // create a new node
    qNode<T> *newNode = new_node(newData);

    // inserting into an empty queue
    //  how do we know if the queue is empty?
    //  check if the queue is empty
    if ((*frontPtr) == nullptr && (*backPtr) == nullptr)
    {

        // assign the frontPtr to the newNode
        (*frontPtr) = newNode;

        // assign the backPtr to the newNode
        (*backPtr) = newNode;
        return;
    }
```

Analysis on Array Stack implementation

In the verification of the array stack implementation, we need to push() each of the numerical numbers into a stack array variable. The verification process is when all of the numbers have been pushed into an array variable, we pop the top and store that popped value into another stack array variable. The verification process will be verified through when the 1st stack array value is equal to 2nd stack array value. It justifies that top and the bottom, as well as the middle has the same value as how it was popped into the second stack array variable.
Analysis on Linked List Queue implementation
IN the verification of the Linked list Queue, We first enqueue all of the numbers, same as the concept for the array stack implementation, and put it into a temporary node playlist. Once the linked list enqueue is

Customer Numbers	Valid?	Output (Console)	Remark
101 102 103 102 101	Yes		This Customer numbers are Palindromes
101 102 103	No		
205 205	Yes		
501	Yes		
10 20 30 40 50	No		
11 22 33 22 11	Yes		This Customer numbers are Palindromes

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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